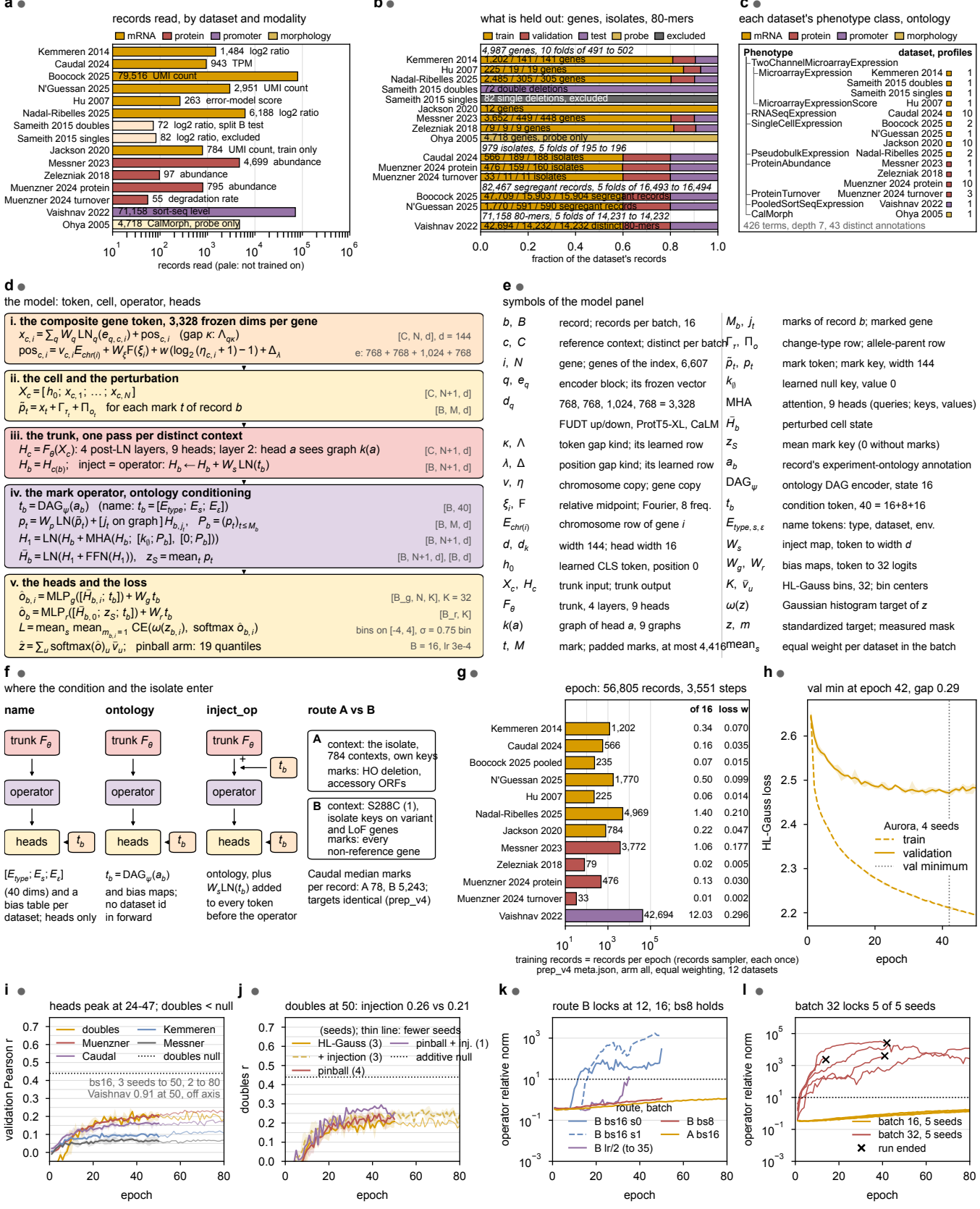

a
records read, by dataset and modality
mRNA protein promoter morphology
Kemmeren 2014 1,484 log2 ratio
Caudal 2024 943 TPM
Boocock 2025 79,516 UMI count
N'Guessan 2025 2,951 UMI count
Hu 2007 263 error-model score
Nadal-Ribelles 2025 6,188 log2 ratio
Sameith 2015 doubles 72 log2 ratio, split B test
Sameith 2015 singles 82 log2 ratio, excluded
Jackson 2020 784 UMI count, train only
Messner 2023 4,699 abundance
Zelezniak 2018 97 abundance
Muenzner 2024 protein 795 abundance
Muenzner 2024 turnover 55 degradation rate
Vaishnav 2022 71,158 sort-seq level
Ohya 2005 4,718 CalMorph, probe only
records read (pale: not trained on)
b
what is held out: genes, isolates, 80-mers
train validation test probe excluded
4,987 genes, 10 folds of 491 to 502
Kemmeren 2014 1,202 / 141 / 141 genes
Hu 2007 225 / 19 / 19 genes
Nadal-Ribelles 2025 2,485 / 305 / 305 genes
Sameith 2015 doubles 72 double deletions
Sameith 2015 singles 82 single deletions, excluded
Jackson 2020 12 genes
Messner 2023 3,652 / 449 / 448 genes
Zelezniak 2018 79 / 9 / 9 genes
Ohya 2005 4,718 genes, probe only
979 isolates, 5 folds of 195 to 196
Caudal 2024 566 / 189 / 188 isolates
Muenzner 2024 protein 476 / 159 / 160 isolates
Muenzner 2024 turnover 33 / 11 / 11 isolates
82,467 segregant records, 5 folds of 16,493 to 16,494
Boocock 2025 47,709 / 15,903 / 15,904 segregant records
N'Guessan 2025 1,770 / 591 / 590 segregant records
71,158 80-mers, 5 folds of 14,231 to 14,232
Vaishnav 2022 42,694 / 14,232 / 14,232 distinct 80-mers
fraction of the dataset's records
c
each dataset's phenotype class, ontology
mRNA protein promoter morphology
Phenotype
dataset, profiles
TwoChannelMicroarrayExpression
MicroarrayExpression
Kemmeren 2014 1
Sameith 2015 doubles 1
Sameith 2015 singles 1
MicroarrayExpressionScore
Hu 2007 1
RNASeqExpression
Caudal 2024 10
SingleCellExpression
Boocock 2025 2
N'Guessan 2025 1
Jackson 2020 10
PseudobulkExpression
Nadal-Ribelles 2025 2
ProteinAbundance
Messner 2023 1
Zelezniak 2018 1
Muenzner 2024 protein 10
ProteinTurnover
Muenzner 2024 turnover 3
PooledSortSeqExpression
Vaishnav 2022 1
CalMorph
Ohya 2005 1
426 terms, depth 7, 43 distinct annotations
d
the model: token, cell, operator, heads
i. the composite gene token, 3,328 frozen dims per gene
[C, N, d], d = 144
e: 768 + 768 + 1,024 + 768
ii. the cell and the perturbation
for each mark t of record b
[C, N+1, d]
[B, M, d]
iii. the trunk, one pass per distinct context
inject = operator:
[C, N+1, d]
[B, N+1, d]
iv. the mark operator, ontology conditioning
(name: )
[B, 40]
[B, M, d]
[B, N+1, d]
[B, N+1, d], [B, d]
v. the heads and the loss
[B_g, N, K], K = 32
[B_r, K]
bins on [-4, 4], σ = 0.75 bin
pinball arm: 19 quantiles
B = 16, lr 3e-4
e
symbols of the model panel
record; records per batch, 16
reference context; distinct per batch
gene; genes of the index, 6,607
encoder block; its frozen vector
768, 768, 1,024, 768 = 3,328
FUDT up/down, ProtT5-XL, CaLM
token gap kind; its learned row
position gap kind; its learned row
chromosome copy; gene copy
relative midpoint; Fourier, 8 freq.
chromosome row of gene i
width 144; head width 16
learned CLS token, position 0
trunk input; trunk output
trunk, 4 layers, 9 heads
graph of head a, 9 graphs
mark; padded marks, at most 4,416
marks of record b; marked gene
change-type row; allele-parent row
mark token; mark key, width 144
learned null key, value 0
MHA
attention, 9 heads (queries; keys, values)
perturbed cell state
mean mark key (0 without marks)
record's experiment-ontology annotation
ontology DAG encoder, state 16
condition token, 40 = 16+8+16
name tokens: type, dataset, env.
inject map, token to width d
bias maps, token to 32 logits
HL-Gauss bins, 32; bin centers
Gaussian histogram target of z
standardized target; measured mask
equal weight per dataset in the batch
f
where the condition and the isolate enter
name
ontology
inject_op
route A vs B
trunk
operator
heads
A context: the isolate, 784 contexts, own keys marks: HO deletion, accessory ORFs
B context: S288C (1), isolate keys on variant and LoF genes marks: every non-reference gene
(40 dims) and a bias table per dataset; heads only
and bias maps; no dataset id in forward
ontology, plus added to every token before the operator
Caudal median marks per record: A 78, B 5,243; targets identical (prep_v4)
g
epoch: 56,805 records, 3,551 steps
of 16
loss w
Kemmeren 2014 1,202 0.34 0.070
Caudal 2024 566 0.16 0.035
Boocock 2025 pooled 235 0.07 0.015
N'Guessan 2025 1,770 0.50 0.099
Hu 2007 225 0.06 0.014
Nadal-Ribelles 2025 4,969 1.40 0.210
Jackson 2020 784 0.22 0.047
Messner 2023 3,772 1.06 0.177
Zelezniak 2018 79 0.02 0.005
Muenzner 2024 protein 476 0.13 0.030
Muenzner 2024 turnover 33 0.01 0.002
Vaishnav 2022 42,694 12.03 0.296
training records = records per epoch (records sampler, each once)
prep_v4 meta.json, arm all, equal weighting, 12 datasets
h
val min at epoch 42, gap 0.29
HL-Gauss loss
Aurora, 4 seeds
train
validation
val minimum
epoch
i
heads peak at 24-47; doubles < null
validation Pearson r
doubles
Kemmeren
Muenzner
Messner
Caudal
doubles null
bs16, 3 seeds to 50, 2 to 80
Vaishnav 0.91 at 50, off axis
epoch
j
doubles at 50: injection 0.26 vs 0.21
(seeds); thin line: fewer seeds
HL-Gauss (3)
pinball + inj. (1)
+ injection (3)
additive null
pinball (4)
doubles r
epoch
k
route B locks at 12, 16; bs8 holds
operator relative norm
route, batch
B bs16 s0
B bs8
B bs16 s1
A bs16
B lr/2 (to 35)
epoch
l
batch 32 locks 5 of 5 seeds
operator relative norm
batch 16, 5 seeds
batch 32, 5 seeds
run ended
epoch